5.2 Strong Induction

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Strong induction is similar to what we've been doing but with a "stronger" assumption (inductive hypothesis). Rather than only assuming $P(k)$ is true, we assume $P(k), P(k-1), P(k-2), \ldots, P(\text{base})$ are <u>all</u> true. This gives us more expressions to use if we get stuck in the "Notice" step.

Prove that if a_n is defined by $a_0 = 1$, $a_1 = 3$, $a_n = 2a_{n-1} - a_{n-2}$ for $n \geq 2$, then for all non-negative integers n, $a_n = 2n+1$.

Let a_n be defined by $a_0 = 1$, $a_1 = 3$ and $a_n = 2a_{n-1} - a_{n-2}$ for $n \geq 2$.

Let $P(n)$: "$a_n = 2n+1$" where n is any non-negative integer.

<u>Base Case ($n=0$)</u>: $P(0)$: "$a_0 = 2 \cdot 0 + 1$" where $2 \cdot 0 + 1 = 1$ so $P(0)$ is true.

<u>Strong Inductive Step</u>: Assume $P(j)$: "$a_j = 2j+1$" for every $0 \leq j \leq k$ for some arbitrary integer $k \geq 0$. We want to show that $P(k+1)$: "$a_{k+1} = 2(k+1)+1$" is true. Notice

$$a_{k+1} = 2a_k - a_{k-1} = 2(2k+1) - (2(k-1)+1)$$

by $P(k)$ and $P(k-1)$. Then

$$2(2k+1) - (2(k-1)+1) = 4k+2 - (2k-2+1)$$

$$= 4k+2 - 2k+1 = 2k+2+1 = 2(k+1)+1.$$

Then $a_{k+1} = 2(k+1)+1$, so $P(k+1)$ is true.

$= 4k+2 - 2k+1 = 2k+2+1 = 2(k+1)+1.$

Then $a_{k+1} = 2(k+1)+1$, so $P(k+1)$ is true.

Therefore by mathematical induction if a_n is defined as $a_0 = 1$ $a_1 = 3$ $a_n = 2n+1$ for $n \geq 2$ then $a_n = 2n+1$ for any non-negative integer n $\blacksquare$

Do we need weak or strong induction to prove $12 \mid n^4 - n^2$ for any positive integer n?

What goes wrong with "weak" induction?

We assume $12 \mid k^4 - k^2$

or $k^4 - k^2 = 12\ell$ for some $\ell \in \mathbb{Z}$.

Then what about $k+1$?

$$(k+1)^4 - (k+1)^2 = k^4 + 4k^3 + 6k^2 + 4k + 1 - (k^2 + 2k + 1)$$

$$= k^4 - k^2 + 4k^3 + 6k^2 + 2k$$

$$= 12\ell + 4k^3 + 6k^2 + 2k$$

can't divide 12 out!

Need more information/statements to use. Likely requires strong induction

The argument form for strong induction is

$$\left[P(0) \wedge \forall k \left(P(0) \wedge P(1) \wedge \cdots \wedge P(k) \Rightarrow P(k+1)\right)\right] \Rightarrow \forall n\, P(n)$$

assuming the base case is $n=0$.

Match Game: Two piles of some number of matches. Two players take turns removing a positive number of matches from either pile. The player who removes the last match wins.

Prove that if both piles have the same number who removes the last match wins.

Prove that if both piles have the same number of matches, Player 2 can always win the match game.

Let $P(n)$: "Player 2 can win if there are n matches in each pile" where n is a positive integer.

Base Case ($n=1$): $P(1)$: "Player 2 can win if there is 1 match in each pile." In this case, Player 1 can only take 1 match and leave 1 behind. Player 2 takes the remaining match and wins so $P(1)$ is true.

Strong Inductive Step: Assume $P(j)$: "Player 2 can win if there are j matches in each pile" for every $1 \leq j \leq k$ for some arbitrary positive integer k. We want to show $P(k+1)$: "Player 2 can win if there are $k+1$ matches in each pile" is true. If the match game starts with $k+1$ matches in each pile, Player 1 must go first. They must take some number $1 \leq x \leq k+1$ matches from either pile. If Player 1 takes all $k+1$ matches from a pile, Player 2 takes all $k+1$ matches from the other pile and wins. If Player 1 takes x many matches from a pile, then Player 2 can take x many matches from the other pile. This leaves exactly $(k+1)-x \leq k$ many matches in both piles so Player 2 can win by $P(k+1-x)$.

Therefore if both piles contain the same number of matches, Player 2 can win the match game ■